METRIC

MIL-C-63982A (AR)
AMENDMENT 1
29 October 1994
SUPERSEDING
SPECIFICATION
26 August 1992

MILITARY SPECIFICATION

CARTRIDGE, 30MM HIGH EXPLOSIVE DUAL PURPOSE, M789

This amendment forms a part of MIL-C-63982A (AR), dated 26 August 1992, and is approved for use by the US Army Armament Research, Development and Engineering Center, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 2

2.1.2: Delete "9390627 - Barrel, Test, 30mm (ADEN Barrel)"

PAGE 4

3.8: Delete second sentence and substitute "The standard deviation shall not exceed 15 m/s at temperatures of 21^0C + 5^0C and 71^0C ± 500. The standard deviation shall not exceed 20 m/s at a temperature of -54^0C + 5^0C.

3.9: Delete in its entirety and substitute the following:

"3.9 Case mouth pressure. In the M230 test barrel (Dwg 9390748) the mean peak pressure shall not exceed 335 MPa at 21^oC ± 5^oC. The mean peak pressure shall not exceed 430 MPa at temperatures over the range from -54^oC to 71^oC. Following the exposure to environmental conditions of paragraph 3.10, the mean peak pressure shall not increase nor decrease from the mean peak pressure at 21^0C + 5^0C by more than 60 MPa."

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PAGE 5

3.11: Delete test barrel dwg. number and Substitute "9390748”

3.15: Delete in its entirety and substitute the fOllowing:

3.15 Normal impact function. When fired against 1.6 mm (nominal thickness) 2024-T3 aluminum alloy sheet, the projectile upon impact shall produce an average hole of not less than seven (7) inches in diameter exclusive of petals, and show evidence of shrapnel on the forward face or remnants of the aluminum sheet."

PAGE 9

Table I, under components:

Delete: "Projectile, 30mm, HEDP, M789 Dwg. 11825955)

Examination for defects	3232	32	ACC-Q REJ-1	3.1	4.4.2.9"

"Projectile, 30mm, HEDP, V789 Dwq. 11825955)

Examination for defects Fuze Torque				3.1.2	4.4.2.9”

PAGE 12

4.4.1: Delete in its entirety and substitute the following:

"4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078. Unless otherwise specified, component parts shall be homogeneous and of a quantity convenient o the contractor except that each cartridge lot shall contain:

a. Cartridge cases (Dwg 11825902) from one unchanged process, from one lot interfix number and one manufacturer.

b. Projectile metal parts (Dwg 9349963) from one unchanged process, from one lot interfix number with not more than two (2) lot numbers from one manufacturer.

c. Primers (Dwg 11825910) from one unchanged process, from one lot interfix number with not more than two (2) lot numbers from one manufacturer.

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PAGE 12 (continued)

4.4.1 (continued)

d. Propellant (Dwg 11825946) from one lot number from one manufacturer.

e. Fuzes (Dwg 11834312) from one unchanged process, from one lot interfix number and one manufacturer."

PAGE 22

4.4.2.7, Major 101: Delete "Visual/weight 1/" and substitute "Weight/Gage 1/".

PAGE 24

4.4.2.9: Add:

"104 Presence of thread lock **Level II** 3.1.2 Gage 1/"

To note block add: "1/ Presence of thread lock shall be checked using automated methods prior to assembly of fuze to charged body."

PAGE 34

4.5.1: Delete test barrel dwg. number and substitute "9390748".

PAGE 35

4.5.4: Delete last sentence and substitute the following:

"Any corrected action time of 1.92 milliseconds or less shall be disregarded as instrumentation error."

4.5.6: Delete in its entirety and substitute the following:

"4.5.6 Projectile structural integrity. Projectile structural integrity shall be verified in conjunction with the ballistic tests of 4.5.2, 4.5.3 and 4.5.4. A witness board/card shall be located approximately 5 meters from the gun muzzle and a witness box shall be located at the muzzle and extend at least one meter forward from the muzzle. The witness box shall be open at each end and be lined with ½ inch thick Celotex."

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PAGE 35 (continued)

4.5.7: Delete in its entirety and substitute the following:

"4.5.7 Environmental tests Unless otherwise specified, performance after environmental exposure, as specified in the applicable test procedure, shall consist of conducting the tests of 4.5.2, 4.5.3, 4.5.4 and 4.5.14.1, except all test rounds shall be conditioned at 21^0C ± 500 prior to firing. Seventy-five percent (75%) of each environmental test sample shall be designated for test 4.5.2, 4.5.3 and 4.5.4 and the balance (25%) designated for test 4.5.14.1."

PAGE 36

Figure 1: Delete in its entirety and substitute with attached Figure 1.

PAGE 38

4.5.7.3: Delete in its entirety and substitute the following:

"4.5.7.3 Aircraft vibration. The ammunition shall be subjected to a vibration profile in accordance with the requirements of MIL-STD-810, Method 514.2, Procedure I, Category C. The test will b conducted with an M592 Ammo Can. The vibration spectrum shall be as shown in Figures 1 and 2. The test duration shall be 45 minutes in each of three mutually perpendicular axis at each of two temperature conditions, -40 ± 2.8°C and 65.5 ± 2.8°C. Following exposure, the test items shall be subjected to the tests referenced in 4.5.7."

PAGE 40

4.5.15: Delete in its entirety and substitute the following:

"4.5.15 Normal impact function. The target shall be placed 200 meters ::: 20 meters from the muzzle. Failure of any projectile to function high order detonation and produce a hole as defined in paragraph 3.15 shall be considered a defect. Failure of 11 or more projectiles shall be cause for rejection of the lot. If 3 to 10 failures occur, testing will continue in accordance with the following double sampling plan:

nl-90	cl=2	rl=1l
nt-270	ct=l0	rt=ll

If two (2) or less failures occur during the initial test sample, the lot will be accepted."

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5.3: After "...charge weights, ~ add 'fuze and lot identification.'

PAGE 45

* 6.14: Delete "SMCAR-QAF-I(D)" and substitute AMSTA-AR-QAC-C.

Custodian:
Army –AR

Preparing activity:
Army-AR

(Project 1305-0102)

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